

**When AI Meets the Public:
How AI-Related Social Media Crises Challenge
Existing Trust & Safety Frameworks**

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Abstract

Trust and Safety frameworks rest on a foundational assumption that harmful content is created by identifiable human agents and hosted on platforms acting as intermediaries. Generative AI disrupts both sides of that assumption simultaneously. The author analyzed 26 AI-related social media crisis events from November 2022 through April 2026 by mapping each onto a four-axis framework that combines TSPA Abuse Types with three non-content trust and safety axes (behavioral and authenticity-based enforcement, recommendation-system intervention, and design-level safety). Sixteen of 26 events (62%) presented structural misfits in which the framework's foundational premises themselves did not hold; the proximate actor was a Platform-Generated or Hybrid-Agency configuration in 13 events, outpacing the User-Originated structure that legacy frameworks foreground. Three recurring structural patterns emerged: attribution ambiguity, the collapse of the content-tool distinction, and design-layer harms with no content-level analog. The paper proposes five policy recommendations responsive to those patterns. The dataset and a working prototype of the proposed dual-coding diagnostic flag are published as a companion artifact at <https://tspa-ai-mapping.vercel.app>, where readers can inspect the per-event coding and exercise the proposed extension to the TSPA taxonomy directly.

I. Introduction

Since the public release of ChatGPT in November 2022, generative AI tools have become infrastructure for everyday content production on the social media platforms the trust and safety field has spent two decades learning to govern. By 2026, these tools are no longer external. Meta AI lives across Facebook, Instagram, and WhatsApp (Meta 2023; Meta 2025). Grok ships with X (xAI 2023). TikTok's Symphony stack underwrites a growing share of advertiser content (TikTok 2024a). My AI sits inside Snapchat alongside human friends (Snap 2023a).

This paper takes the proliferation of AI-related social media crises since November 2022 as its empirical object and asks whether the classification frameworks the trust and safety field has developed, the TSPA Abuse Types taxonomy chief among them, still describe the phenomena they were built to describe. They do not. The misfit, however, is not a matter of missing categories. Existing frameworks rest on a foundational triangle: a human agent creates content, a platform serves as intermediary, and trust and safety professionals manage the relationship between them. Generative AI disrupts both the content-side and intermediary-side corners of that triangle. The creator may be a model or a hybrid human-machine workflow rather than an identifiable human. The intermediary may be the AI provider that produced the content, not an external party hosting it. When AI-related crises since 2022 are mapped against the TSPA taxonomy and the broader non-content trust and safety toolset, the resulting misfits cluster into recurring patterns that point to structural challenges in the framework's foundational premises about agency, attribution, and the nature of moderated content.

The literature on AI safety and the literature on platform trust and safety have developed largely in parallel: the former asks what models can produce, the latter how human-generated content is governed once posted. Neither field has systematically examined how AI-related incidents originate, escalate into social media crises, and remain legible or illegible to the classification tools practitioners use day to day. This paper situates itself in that gap.

The contribution is threefold. The paper first assembles an event-level record of 26 AI-related social media crises from November 2022 through April 2026 from international news coverage. It then maps each event onto the TSPA Abuse Types taxonomy and against three additional axes the trust and safety field already deploys, behavioral and authenticity-based enforcement, recommendation-system intervention, and design-level safety, to test where existing classifications succeed, partially succeed, or fail. Finally, it distinguishes patchable gaps from structural misfits and proposes five policy recommendations responsive to the structural challenges, each accompanied by an explicit assessment of implementation hurdles.

Two qualifications bound the argument. Industry threat reports from Meta, OpenAI, Anthropic, and Google find that generative tools have so far yielded productivity gains for adversaries more often than wholesale evasive capability (OpenAI 2024; Anthropic 2025; Google Threat Intelligence Group 2025). Platform regulation, meanwhile, has not remained content-centric: the Digital Services Act, the UK Online Safety Act, and the Australian eSafety Commissioner's Safety-by-Design framework already direct governance toward systemic risk and recommender accountability (European Union 2022; Ofcom 2024; eSafety Commissioner 2021). The argument here is correspondingly narrower. Even where regulation has moved beyond content, it still presumes a platform institutionally separable from the harmful actor. Once the platform is itself the AI provider, that separation is the assumption that fails first.

The empirical analysis below is informed by the author's parallel construction of the dataset described in Section III, hand-coded against the framework axes specified there. A companion artifact at <https://tspa-ai-mapping.vercel.app> makes the dataset and the author's framework-mapping coding inspectable on a per-event basis, and provides a working prototype of the diagnostic-flag policy recommendation developed in Section VI.D. The dataset and the prototype are integral to the analysis rather than supplementary to it; they make the methodology and one of the policy recommendations concrete in a form a reader can verify directly.

II. Literature Review

A. Trust and Safety Frameworks and Their Premises

Modern trust and safety practice is structured around content moderation in its narrow sense and its non-content extensions. On the content side, the Trust and Safety Professional Association's Abuse Types taxonomy provides the field's shared vocabulary, organizing harms under recurring categories such as Hateful Content, Harassment, Dangerous Misinformation, Impersonation, and Fraud, framed as "underlying human misbehaviors" (TSPA n.d.). Eric Goldman's "Content Moderation Remedies" (2021) extends the response-side vocabulary, cataloging a wide menu of interventions beyond removal: visibility reduction, distribution penalties, monetization restrictions, friction, and account-level changes. Both frameworks share a structural assumption: the regulatory object is a user's online content or actions, redressed by a service that sits above and outside it.

The trust and safety field has not remained at the content layer. Its non-content tools cluster around three additional axes that the present paper takes seriously rather than caricaturing. The first is behavioral and authenticity-based enforcement, exemplified by Camille François's "actors, behaviors, content" framework (François 2019) and operationalized in Meta's Coordinated Inauthentic Behavior policy (Meta 2024a), in which deceptive campaigns are regulated through

behavioral signals even when individual posts do not violate content rules. The second is recommendation-system intervention. Daphne Keller's "Amplification and Its Discontents" (Keller 2021) frames the doctrinal turn: platforms may not be liable for what users say, but they may be regulated for what they amplify (see also Gillespie 2022). The third is design-level safety, descended from Cavoukian's Privacy by Design (Cavoukian 2011) and codified in the Australian eSafety Commissioner's Safety by Design principles (eSafety Commissioner 2021); the provenance ecosystem, C2PA credentials and watermarking systems such as SynthID, belongs to this lineage (C2PA 2025; Google DeepMind 2024). These three axes meaningfully exceed content classification but do not exceed the framework's deeper premise. Behavioral enforcement still presupposes that "inauthentic behavior" traces to manipulative actors whose identities are merely concealed. Recommendation governance still presupposes a platform-intermediary that decides whether to amplify others' content. Design-level safety still presupposes designers who can anticipate misuse pathways. The shared foundational assumption is the legibility of human agency: harmful action originates with people whose identities, behaviors, and intentions can in principle be inferred.

B. AI Safety and Governance

The literature on AI safety has developed in a different idiom. Its dominant concerns are model-level: alignment, red-teaming, refusal training, capability evaluations, and responsible deployment (NIST 2023). The object of governance is the model and its outputs in controlled testing, not the social system into which the outputs eventually flow. The regulatory landscape mirrors this orientation. The EU Artificial Intelligence Act regulates AI systems by risk tier and imposes specific transparency obligations on generative content under Article 50 (European Union 2024). The TAKE IT DOWN Act in the United States and several state-level deepfake statutes target specific harms produced through AI rather than the platform context in which those harms circulate. Where platform regulation has approached AI directly, it has typically done so through systemic-risk vocabulary already developed for non-AI harms. The Digital Services Act's Articles 27, 34, and 38 require very large online platforms to assess systemic risks "stemming from the design and functioning of their service, including algorithmic systems" and to provide non-profiling-based recommender options (European Union 2022). The UK Online Safety Act and Ofcom's implementing guidance impose analogous duties (Ofcom 2024).

This is a significant movement, and one this paper takes seriously. It is also a movement in which AI is largely an additional risk vector folded into pre-existing systemic-risk regimes, rather than a phenomenon that asks whether the underlying premises of those regimes still hold.

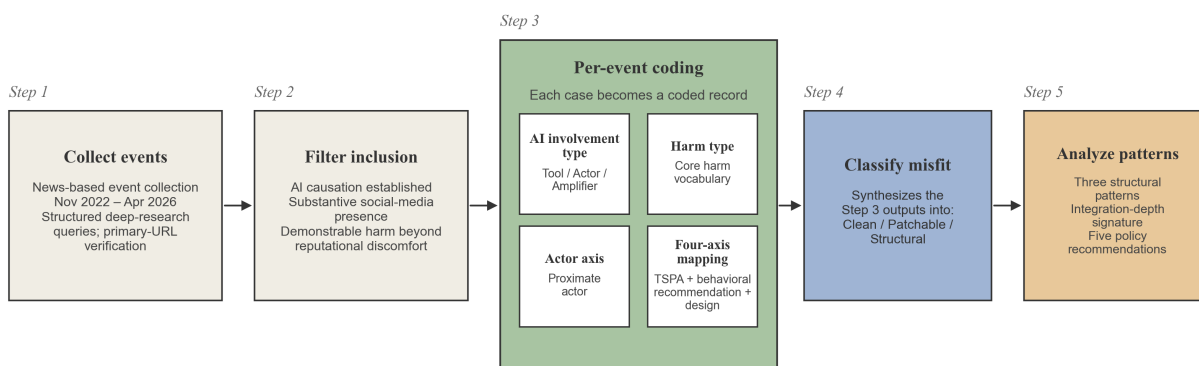
C. The Gap This Paper Addresses

What neither field has systematically examined is the connective tissue between them: how AI-related incidents originate, escalate into social media crises, and become legible (or remain illegible) to the classification tools trust and safety practitioners use every day. AI safety research stops where the model output ends. Platform trust and safety research begins with content already on the platform and an actor already accountable for it. The intermediate phase, where the model, the user, the platform, the recommender, and the design surface co-produce something that may or may not be classifiable, has no consolidated literature. Event-level analysis, in which discrete

incidents are examined for how they originate, propagate, and resolve, provides a methodological lens suited to that intermediate phase. The next section describes how this paper applies it.

III. Methodology

The methodology proceeds in five steps grouped into two stages: data collection (Steps 1–2) to map the landscape of AI-related social media crises, then framework mapping (Steps 3–5) to test where existing trust and safety classifications succeed and fail. The second stage is the paper's core analytical contribution.



N = 26 events. Full per-event coding published at <https://tspa-ai-mapping.vercel.app>

Figure 1. Methodology overview. The study proceeds in five steps from news-based event collection through inclusion filtering, per-event coding across the four-axis framework, misfit classification, and pattern analysis. The full per-event coding is published at the companion artifact.

A. Data Collection

The author constructed an event-level dataset of 26 AI-related social media crisis events from November 2022 through April 2026, drawn from international news coverage. News sources were used because they provide external validation that an event achieved societal significance and because their narrative structure permits systematic identification of stakeholders, platforms, and outcomes. An event qualified for inclusion if generative AI played a central role in causing, enabling, or amplifying harm; if it had substantive presence on one or more social media platforms; and if it produced demonstrable harm beyond reputational discomfort.

Initial collection used keyword-based queries combining AI terminology (artificial intelligence, generative, deepfake, synthetic media, large language model) with social media and platform terms. Two structured deep-research queries assisted source aggregation; the queries are reproduced in Appendix B. All sources surfaced were independently verified by the author through primary URLs and publication checks before inclusion. Sources whose AI-related causation could not be established from primary reporting were excluded.

B. Event Classification

Each event is coded along four dimensions designed to capture features distinctive to AI-related crises. Table 1 specifies the dimensions and the values each can take.

Table 1. Coding dimensions for the 26-event dataset, with the values each dimension records.

Dimension	What it captures
AI involvement type	Tool (humans using AI to cause harm), Actor (AI system failure causing unintended harm), or Amplifier (AI scaling pre-existing harmful dynamics)
Stakeholders	Who caused or enabled the harm, who was harmed, and who responded, drawn from platform, AI developer, government, and civil-society actors
Harm type and platform(s)	Primary harm category (per the TSPA taxonomy) and the social-media platform or platforms on which the event circulated
Response and outcome	Actions taken in response to the event and the actor that took them; resolution status as of the data-collection cutoff

C. Framework Mapping

The framework-mapping coding was performed by the author across all 26 events. Each event was coded along two fields per axis (a fit status and a written rationale) across four axes (TSPA Abuse Types, behavioral and authenticity-based enforcement, recommendation-system intervention, design-level safety), plus a final misfit classification (clean / patchable / structural) and a rationale for that classification. The full per-event coding is published at the companion artifact (Appendix C) and can be inspected on a per-event basis at <https://tspa-ai-mapping.vercel.app>.

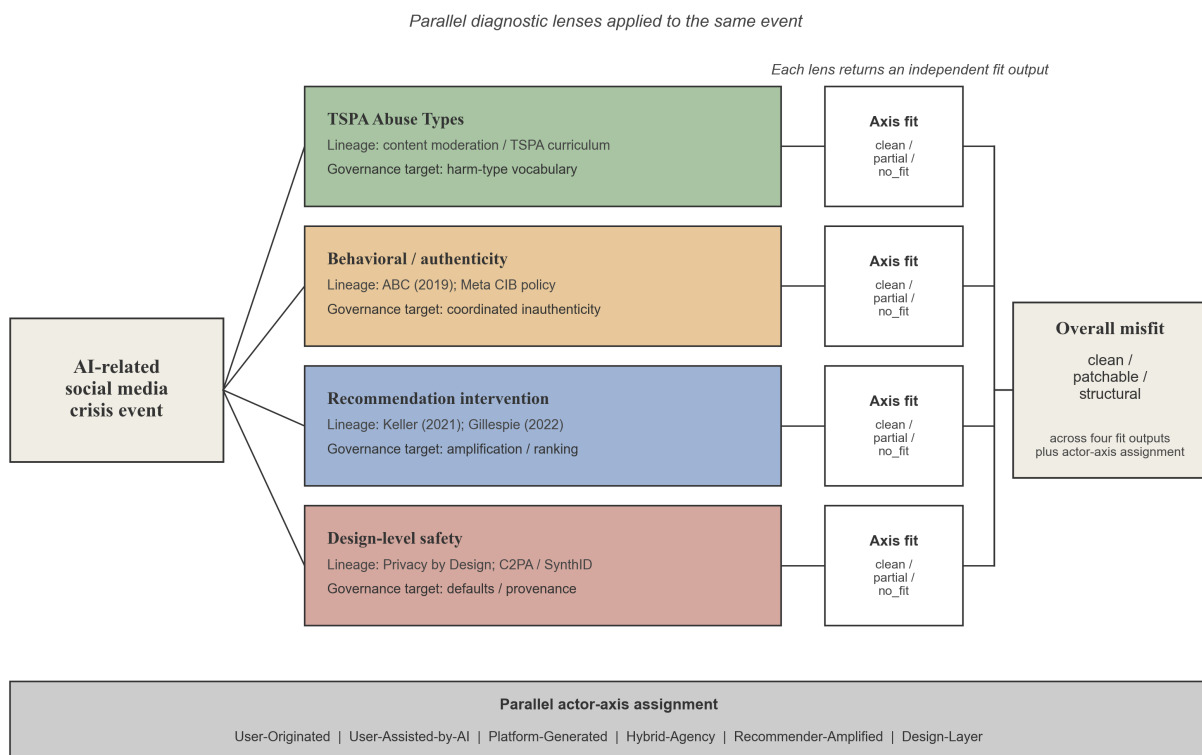


Figure 2. Four-axis framework architecture. Each AI-related social media crisis event is coded against TSPA Abuse Types and three non-content axes (behavioral and authenticity-based enforcement, recommendation-system governance, design-level safety). Each axis fit (clean / partial / no_fit) feeds into an overall misfit classification (clean / patchable / structural).

The TSPA Abuse Types taxonomy serves as the primary reference because it functions as the field's shared vocabulary. The three non-content axes ensure that any framework misfit identified is not an artifact of testing only the content layer. For each event the mapping documents whether the event fits cleanly into existing categories, partially fits with information loss, or resists classification entirely. For partial and non-fitting cases, the analysis identifies the specific reason for misfit and classifies it as either *patchable* (the framework lacks a subcategory but its underlying assumptions remain intact) or *structural* (the misfit follows from a foundational assumption the framework cannot accommodate without revision). This distinction does significant analytical work in this paper because the appropriate field response differs sharply by category.

IV. Findings

A. The Landscape

Across the November 2022 through April 2026 window, the dataset surfaces several visible patterns. AI-related crises distribute across all major social media platforms, with concentrations on X (11 events), Meta-owned services (7 events), TikTok (4 events), and Snap (3 events), with several events spanning multiple platforms. This distribution reflects both audience size and the integration of generative tools into those platforms' product surfaces. The temporal distribution shows three approximate phases. An initial 2023 phase is dominated by external-tool harms: deepfakes circulated on platforms, ChatGPT outputs used to generate misinformation. A 2024 phase shows influence operations increasingly using generative tooling for persona maintenance and localization (OpenAI 2024; Anthropic 2025). A 2025 to 2026 phase shows platform-integrated AI features themselves becoming sources of crisis (Horwitz 2025a; Horwitz 2025b; Singh 2025).

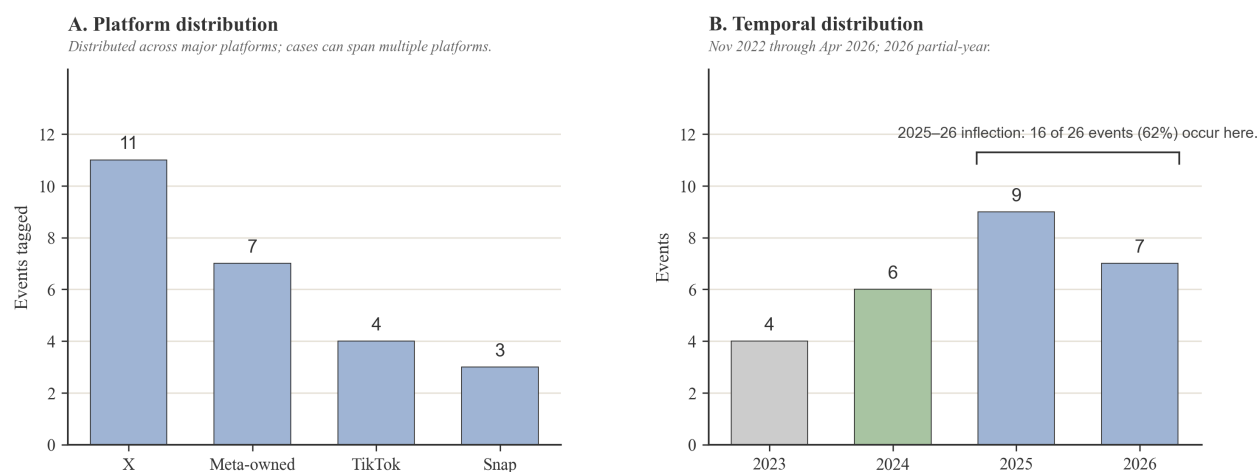


Figure 3. Landscape of 26 AI-related social media crisis events (November 2022 – April 2026). (A) Distribution across major platforms, with several events tagged on multiple platforms. (B) Temporal distribution by year, indicating intensification of platform-integrated AI cases in 2025.

The harm types span the familiar: dangerous misinformation, hateful content, harassment, sexualized abuse, fraud, impersonation, child safety. What changes is not the harm vocabulary but the locus of agency. By 2025 the dataset includes incidents in which the immediate producer of harm is not a discrete user but a platform-integrated assistant, an AI persona enabled by a creator-tooling layer, or an automated ad-generation system that operates without clear user authorship.

B. Framework Mapping Results

Of the 26 events coded for this study, 3 (12%) fit cleanly into existing frameworks at both the harm-type and actor-axis layers; 7 (27%) presented patchable misfits, where harm-type categories applied at the surface but information was lost at the actor or design layer in ways a subcategory addition could repair; and 16 (62%) presented structural misfits, where the framework's foundational premises themselves did not hold and revision rather than extension was required. The actor-axis distribution further surfaces the integration trend: across all 26 events, the proximate actor was a Platform-Generated system in 7 cases and a Hybrid-Agency configuration in 6, together exceeding the 5 cases coded as User-Originated and outpacing Recommender-Amplified (4) and Design-Layer (4) cases as well. The full per-event coding is browseable at the companion artifact at <https://tspa-ai-mapping.vercel.app>, where readers can filter by year, platform, AI involvement type, and misfit classification.

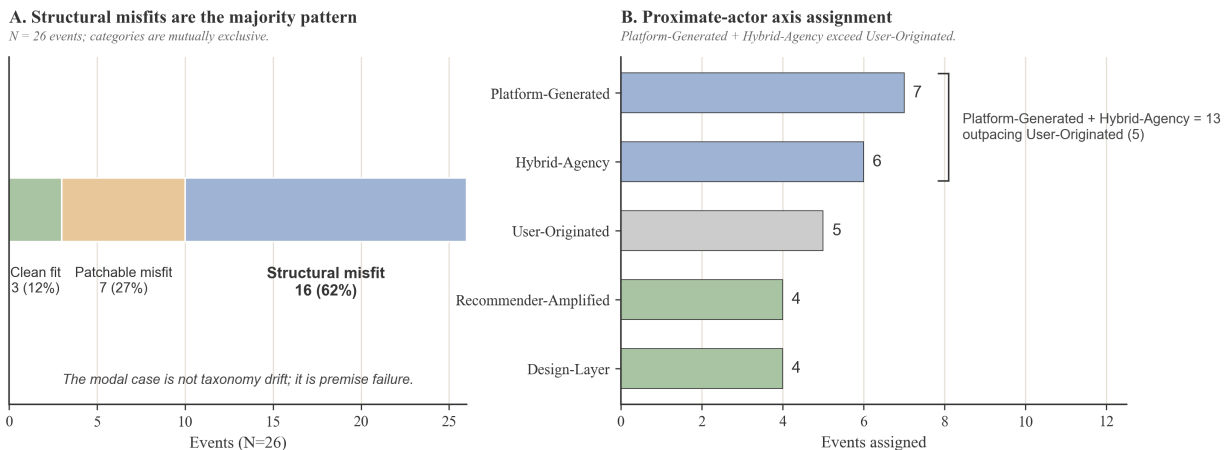


Figure 4. Distribution of framework-mapping outcomes across 26 events. (A) Misfit classification: 16 of 26 events (62%) presented structural misfits where the framework's foundational premises did not hold. (B) Proximate-actor axis: Platform-Generated and Hybrid-Agency assignments together (13 of 26) outpaced User-Originated assignments (5), the actor structure the legacy frameworks foreground.

The non-content axes show a similar pattern. Behavioral enforcement frameworks accommodate AI-assisted persona networks like the Anthropic-documented influence-as-a-service operation only awkwardly, because the model is now orchestrating engagement timing rather than a human operator (Anthropic 2025). Recommendation-governance frameworks accommodate the "All Eyes on Rafah" virality and the 2025 Polish complaint about TikTok-distributed AI content only by treating the recommender as the proximate agent, a move the frameworks permit but do not foreground (Guardian 2024; European Commission 2024; Reuters 2025b). Design-level frameworks accommodate provenance failures, C2PA stripping and watermark removal, only by acknowledging that the design defaults do not survive contact with downstream platform pipelines (AFIP 2025; Zhao et al. 2024).

C. Structural Misfit Patterns

Three recurring structural patterns emerge from the partial- and no-fit cases.

The first is *attribution ambiguity*. When AI is the proximate producer of content, the "giver" presupposed by content-centered moderation is no longer well-defined. The user prompted; the model generated; the platform configured; the recommender amplified. Each contribution is real, and none on its own captures the harm. The mapping cannot be fixed by adding a "user used AI" subcategory because the issue is not a missing label but a missing actor.

The second is *the collapse of the content-tool distinction*. In legacy frameworks, content is the object and the platform's tooling is the surface that hosts or fails to host it. Once the platform's own model is the producer of the content, whether through Meta AI, Grok, an integrated TikTok creative tool, or My AI, the tool is the content. Goldman's remedies framework describes interventions on a user's content; here the substantive remedy is on the platform's product behavior.

The third is *novel harm types with no content-level analog*. Some harms identified in the dataset have no clear referent in TSPA categories because they describe the relational or design layer rather than the content layer. The Meta AI Discover feed incident, in which users shared intimate prompts publicly without grasping the publicness of the act, is a UX-driven privacy harm that maps awkwardly to existing privacy categories (Fowler 2025). Reported AI companion dependency does not map to any TSPA harm category. These are not omissions in the taxonomy; they are events that occur in a design layer the taxonomy does not address.

The patchable-versus-structural distinction maps onto these patterns straightforwardly. Adding subcategories for AI-generated impersonation or AI-assisted fraud is patchable: the framework's premises survive. Accommodating attribution ambiguity, content-tool collapse, and design-layer harms is structural: it requires revising the assumption that harm has an identifiable human actor and that the platform sits above the content rather than producing it.

D. Platform-Integrated Generative AI as a Distinct Pattern

The structural misfit patterns become most visible when generative AI is integrated into the platform itself. Across Meta, X, TikTok, and Snap, the integration takes different product forms but produces a recognizable governance signature.

Meta's integration is the broadest, spanning Facebook, Instagram, WhatsApp, Threads, AI Studio, and the standalone Meta AI app (Meta 2023; Meta 2024b; Meta 2025). The 2025 incident cluster reveals the structural problem: Reuters documented internal policies permitting chatbots to engage minors in "romantic or sensual" conversations and enabling flirty celebrity chatbots without permission (Horwitz 2025a; Horwitz 2025b), while the Discover feed surfaced personal prompts publicly without users grasping the publicness (Fowler 2025). Meta's response, retraining and teen-access restrictions (Singh 2025), is product-governance work, not classical moderation.

Grok's integration centers on X's answer-engine layer (xAI 2023; xAI 2025). The 2024 incident in which five U.S. secretaries of state urged Elon Musk to fix Grok's incorrect election information (Associated Press 2024) is the clearest illustration of platform-integrated AI producing public harm: the harmful "speaker" was not a user but the platform's own assistant. Subsequent inappropriate

Grok outputs (Reuters 2025a) and the European Commission's 2026 DSA proceedings against X for AI-related features (European Commission 2026) reinforce the same shape.

TikTok's integration is creative-commercial: Symphony, Digital Avatars, AI Dubbing, and an Ad Assistant embedded in TikTok Ads Manager (TikTok 2024a; Reuters 2024). The Verge documented apparently AI-generated ads lacking the disclosure TikTok's own policies require (Weatherbed 2026), and game publisher Finji alleged that TikTok's automated ad system generated racist and sexually objectifying transformations of its game art despite the company having attempted to disable AI settings, with TikTok support describing the process as "a broader automated initiative" (Francis 2026). Where Meta's incidents involve synthetic agency at the conversational layer, TikTok's involve it at the advertising-automation layer.

Snap places My AI in the chat interface alongside human contacts (Snap 2023a). Testing by the *Washington Post* in 2023 showed My AI offering substance-concealment advice to a purported fifteen-year-old and romantic-encounter advice to a purported thirteen-year-old (Fowler 2023). Snap's response, age signals, parental visibility, restricted access (Snap 2023b), was followed by an Information Commissioner's Office investigation that concluded only after Snap completed a more thorough pre-launch risk assessment (Information Commissioner's Office 2024) and a 2025 Federal Trade Commission referral to the Department of Justice (Federal Trade Commission 2025).

The cross-platform signature is consistent. The platform becomes a multi-role institution: provider of the model, designer of the interface, distributor of the output, and author of the rules that constrain it. The most effective interventions are system-level, retraining, prompt changes, age gates, feature restrictions, pre-launch risk assessment, rather than content-level. Attribution is genuinely unstable: the actor may be the user who prompted, the model that generated, the team that set the system prompt, the product manager who chose the defaults, or the recommender that amplified. The right answer is often "all of the above," which is precisely why the older frameworks misfit.

V. Discussion

A. What the Misfit Patterns Reveal

The misfit patterns identified in Section IV point not to taxonomy completeness but to assumption survivability. The TSPA taxonomy describes the surface harm vocabulary, hateful content, dangerous misinformation, fraud, impersonation, accurately enough that a substantial proportion of dataset events map cleanly. What it cannot describe is who is acting when the proximate actor is hybrid. The non-content axes scaffold the same problem at greater depth: behavioral enforcement assumes manipulative actors whose authenticity can be cross-checked; recommendation governance assumes a platform institutionally separable from the content it amplifies; design-level safety assumes designers who can anticipate misuse pathways from the outside. Each assumption was reasonable for the world in which the framework was developed. Each is destabilized when generative AI is the producer rather than the produced.

B. Patchable Gaps Versus Structural Misfits

The distinction between patchable and structural misfit matters because the field's response should differ. Patchable misfits, missing subcategories for AI-generated impersonation, AI-assisted fraud, AI-driven harassment campaigns, can be addressed with the existing analytic apparatus. They produce taxonomy refinement, training updates, and standard-setting through bodies like TSPA, the Tech Coalition, and the Global Internet Forum to Counter Terrorism. Structural misfits, attribution ambiguity, the collapse of the content-tool distinction, design-layer harms, do not yield to taxonomy refinement. They require revising the assumption that harm has an identifiable human actor or that the platform sits above and outside the content it governs. Refusing this distinction risks expending field energy on patches that do not address what the structural cases need.

C. Connections to Current Regulatory Efforts

Current regulation has begun to move toward system-level governance even before the AI-integration challenge sharpened. The DSA's Articles 27, 34, and 38 already require risk assessment of recommender systems and provision of non-profiling-based options (European Union 2022). The UK Online Safety Act assigns risk-management duties at the service level (Ofcom 2024). The EU AI Act imposes transparency obligations on generative content under Article 50 (European Union 2024). The Information Commissioner's Office's investigation of Snap's My AI explicitly tied lawful deployment to pre-launch risk assessment (Information Commissioner's Office 2024). These movements establish a regulatory theory of platform accountability that need not begin with identification of an offending user.

What they do not yet do is name the structural break that platform-integrated AI introduces. The DSA still presumes that the platform is the system being assessed, not the producer of the harmful content. The ICO's risk-assessment requirement is the closest existing instrument to the conceptual move this paper urges, because it treats the AI feature as a product whose pre-launch assessment is the locus of obligation rather than the post-hoc moderation of its outputs. Generalizing that move is one of the policy directions Section VI develops.

D. Why Frameworks Are Not Enough: The Practitioner Gap

A second line of analysis comes from practitioner perspective. Classroom discussions with industry practitioners during a Spring 2026 trust and safety simulation surfaced a recurring observation: frameworks describe what *can* be done, not what *is* done, and the gap is widening. One former trust and safety leader put it directly: "You can have all the frameworks and policies and guardrails in the world, but ultimately a lot of very important decisions get made because of humans in particular positions with specific incentives." Three more specific observations from the same session inform the analysis here. First, the only generally defensible enforcement lever for AI-amplified influence operations remains coordinated inauthentic behavior, and even this lever is eroding as AI-assisted persona maintenance hollows out authenticity signals. Second, monetization-cessation expectations have proven naïve. Third, the impact-measurement teams whose analyses would underwrite policy responses have been substantially reduced, and existing methodologies cannot keep pace with technological change.

This third observation finds independent support outside the practitioner perspective. Camille François, in April 2026 IGP seminar remarks, framed the same problem in structural rather than

capacity-loss terms: the field lacks a computationally tractable definition of synthetic-content abundance, and "measurement is the first policy barrier" rather than an end-stage difficulty after the policy frame is settled (François 2026). An industry practitioner describing measurement teams as gutted and a leading disinformation researcher describing measurement itself as definitionally incomplete arrive at the same gap through different vantage points.

These observations matter for the present argument because they identify a category of failure the framework analysis alone cannot capture. Even if every misfit identified in Section IV were patched, the frameworks would still not bind. They are necessary conditions for governance, not sufficient ones. The structural misfit is therefore not only conceptual; it is also institutional and evidentiary. This is the practitioner gap.

E. Engaging Counter-Arguments

Three counter-arguments visible in the literature deserve direct engagement. Table 2 organizes them across two analytical dimensions: what each correctly identifies about platform governance today, and how the structural claim survives the constraint each imposes.

Table 2. Three counter-arguments to the structural-misfit claim, with the empirical evidence each marshals and the response from this paper.

Counter-argument	What it correctly identifies	How the structural claim survives
Threat-report consensus. Generative AI has not yet produced wholesale evasive capability (Google Threat Intelligence Group 2025; OpenAI 2024; Meta 2024a).	Industry threat reports document productivity gains for adversaries rather than novel capabilities; the field is not in operational crisis.	The argument is structural, not operational: structural incompatibility is visible in edge cases and platform tooling, and it scales with integration depth (Section IV.D). Threat-report instruments are calibrated to actor-intent operations and exclude intent-absent monetization-pipeline content (François 2026).
Doctrinal evolution beyond content. Daphne Keller (2021), the DSA, and Ofcom's Online Safety Act guidance show platform regulation has moved beyond speaker-level moderation.	System-level governance through risk assessment, recommender accountability, and Article 34 obligations is real and operationally significant.	Even system-level governance presumes a platform institutionally separable from the harmful actor and able to act upon it as intermediary, optimizer, or designer. Platform-integrated generative AI disrupts that separability.
Patchability and institutional adaptation. Some events fit existing categories (TikTok ads, Grok outputs, Meta celebrity chatbots); platforms have begun safety-by-design extensions (Singh 2025; TikTok 2024b; Snap 2023b).	A subset of cases is patchable, and platform safety-by-design adaptation is genuine continuity, not absence of response.	A robust subset of dataset events (Discover feed UX, Finji ad-automation, My AI youth-safety) cannot be patched because the problem is assumption-level. Adaptations are bounded by the same anticipate-misuse premise the design literature itself rests on (Zhao et al. 2024; AFIP 2025).

Taken together, the counter-arguments narrow rather than refute the structural claim. It does not require the immediate operational obsolescence of legacy frameworks. It does require that the field acknowledge an integration-depth dimension along which the frameworks' assumptions become progressively less tenable, and that the structural cases are no longer rare.

VI. Policy Recommendations

The five recommendations below are responsive to the structural misfit patterns identified in Section IV and the discussion in Section V. Each is paired with an empirical basis and an explicit implementation hurdle. They are designed to be additive to the existing regulatory architecture rather than to replace it. Table 3 summarizes the five recommendations in a single view; the subsections that follow develop each in detail.

Table 3. Summary of the five policy recommendations responsive to the structural misfit patterns.

Recommendation	Core proposal: what to require	Empirical basis	Primary implementation hurdle
1. Provenance Preservation Mandate	Require Very Large Online Platforms to preserve C2PA credentials, SynthID watermarks, and equivalent provenance metadata across upload, processing, and re-share pipelines, and to display provenance information to viewers when present.	AFIP (2025): C2PA credentials stripped on 5 of 6 tested platforms; Oversight Board (2026) on Meta surfacing	Watermarks provably removable (Zhao et al. 2024); preservation alone insufficient — requires display obligation plus cryptographic redundancy
2. Pre-Deployment Risk Assessment for Integrated Generative AI	Require platforms launching consumer-facing integrated AI features (assistants, persona marketplaces, automated creative tooling) to complete and publish a pre-launch risk assessment addressing minor safety, hallucination harm, persona disclosure, and downstream content circulation.	ICO (2024) Snap My AI investigation; FTC (2025) referral to DOJ	Speed mismatch: AI product cycles outpace regulatory cycles — requires rolling post-launch reassessment triggers tied to material model updates
3. Synthetic Agency Disclosure Standard	Require persistent, interface-level disclosure when users interact with AI personas, characters, or assistants embedded in a platform's social or conversational surface, reinforced at moments of higher relational risk (medical/legal advice, romantic exchange, minor-facing contexts).	Horwitz (2025a, 2025b) on Meta celebrity chatbots and minor-safety policy; Fowler (2025) on Discover-feed publicness	Friction trade-off: blanket persistent disclosure undercuts utility — requires tiered design with stricter rules for minor-facing and high-risk contexts
4. Hybrid-Agency Taxonomy Extension	Extend the TSPA Abuse Types taxonomy with a parallel actor-axis category (provisionally Platform-Generated / Hybrid-Agency) so that an event can be coded simultaneously as a familiar harm type and as a structurally distinct actor configuration.	Finji (Francis 2026), Discover feed (Fowler 2025), Grok election misinformation (AP 2024) — all recognizable harms with structurally distinct actors	TSPA revision politically slow — phased: introduce a diagnostic flag first, formal taxonomy change after practitioner adoption and case accumulation
5. Recommender Accountability for Synthetic Virality	Require VLOPs' DSA Article 34 systemic-risk assessments to include a synthetic-content amplification metric — the rate at	"All Eyes on Rafah" virality (Guardian 2024); EC (2024) TikTok Romanian	Platform opacity — metric is meaningful only if external auditors can verify it, so the

Recommendation	Core proposal: what to require	Empirical basis	Primary implementation hurdle
	which the platform's recommender foregrounds AI-generated or AI-significantly-altered content relative to organic content, broken down by topic and risk tier, with audit access analogous to DSA Article 40.	election proceedings; Reuters (2025b) Polish complaint on TikTok-distributed AI content	obligation must be coupled with verified-research access provisions

A. Provenance Preservation Mandate

The recommendation is to require Very Large Online Platforms under the DSA, and the equivalent category under the UK Online Safety Act, to preserve C2PA content credentials, SynthID watermarks, and equivalent provenance metadata across upload, processing, and re-share pipelines, with minimum metadata fields that survive normalization and a viewer-facing display obligation when provenance is present.

The basis lies in the AFIP (2025) finding that C2PA manifests were fully stripped on five of six tested platforms and the Oversight Board's 2026 conclusion that Meta's surfacing of AI-generated content "must evolve" (Oversight Board 2026). The recommendation treats provenance as a regulatory object rather than as a voluntary industry adoption.

The primary implementation hurdle is technical. Zhao et al. (2024) demonstrate that invisible watermarks are provably removable using generative regeneration attacks, so preservation alone is insufficient. The mandate should therefore couple metadata preservation with a display obligation, where provenance is preserved, it must be visible to viewers, and pair both with detection redundancy through cryptographic credentials that survive the regeneration class of attacks.

B. Pre-Deployment Risk Assessment for Integrated Generative AI

The recommendation is to require platforms launching consumer-facing integrated generative AI features (conversational assistants, persona marketplaces, automated creative tooling) to complete and publish a risk assessment before launch, addressing minor safety, hallucination-driven harm, persona disclosure, default privacy posture, and downstream content circulation. The assessment should be enforceable by the relevant data-protection or platform-regulation authority and conditioned on ongoing post-launch updates.

The basis is the Information Commissioner's Office's 2024 conclusion of its Snap My AI investigation, which accepted Snap's mitigations only after a more thorough pre-launch risk assessment (Information Commissioner's Office 2024), the most workable existing model. The Federal Trade Commission's 2025 referral of a complaint against Snap to the Department of Justice indicates that the regulatory appetite for treating AI features as products requiring pre-launch scrutiny extends beyond Europe (Federal Trade Commission 2025).

The primary implementation hurdle is the speed mismatch between AI product cycles and regulatory cycles. Pre-launch assessment risks becoming a paper exercise if the underlying model is updated weekly. The recommendation should therefore specify rolling post-launch reassessment

triggered by material model updates, and require disclosure of those triggers, so that the regulatory locus is not a one-time gate but a sustained obligation.

C. Synthetic Agency Disclosure Standard

The recommendation is that when users interact with AI personas, characters, or assistants embedded in a platform's social graph or conversational surface, disclosure should be persistent and interface-level rather than reduced to a one-time label. Persistent disclosure means the synthetic nature of the counterparty is rendered visible across the duration of the interaction and reinforced at moments of higher relational risk (medical and legal advice, romantic or sexual exchange, exchanges involving minors). The standard should be regulator-set and platform-implemented, with stricter requirements in minor-facing contexts.

The basis is Reuters' 2025 finding that Meta-enabled celebrity chatbots claimed to be the real people, including a child celebrity (Horwitz 2025b); the Meta AI internal-policy document permitting romantic conversations with minors (Horwitz 2025a); and the Discover feed incident in which users did not appear to grasp the public nature of their AI interactions (Fowler 2025).

The primary implementation hurdle is friction. Persistent disclosure applied indiscriminately would undercut conversational utility and creator-economy use cases for AI personas. The recommendation therefore specifies a tiered design: minor-facing contexts and high-risk topical zones require strict persistent disclosure; adult, low-risk contexts require initial disclosure with re-asserted disclosure at relational-risk thresholds. The tier boundaries should be regulator-set rather than platform-discretionary.

D. Hybrid-Agency Taxonomy Extension

The recommendation is to extend the TSPA Abuse Types taxonomy with a category that names harms whose proximate actor is the platform itself or the platform's integrated AI system rather than a discrete user. The category, provisionally "Platform-Generated Harm" or "Hybrid-Agency Harm," need not replace existing harm-type categories. It should function as a parallel actor-axis classification applied alongside the standard harm type, so that an event can be coded as Hateful Content (harm) and Platform-Generated (actor) simultaneously.

The basis is the Finji case (Francis 2026), the Meta Discover feed incident (Fowler 2025), and the Grok election misinformation case (Associated Press 2024), each of which is recognizably a familiar harm at the harm-type layer but a structurally distinct event at the actor layer. A working prototype of the proposed dual coding is available at the companion artifact (<https://tspa-ai-mapping.vercel.app>, "Hybrid-Agency Diagnostic Flag" section), where readers can input an event description and apply the harm-axis (TSPA) and actor-axis classifications simultaneously to see what the dual coding makes visible that single-axis coding does not.

The primary implementation hurdle is institutional. TSPA is an industry consortium and taxonomy revision is politically slow. The recommendation therefore proposes a phased approach: a "diagnostic flag" introduced first for analytic and reporting use, with formal category status pursued only after a defined period of practitioner adoption and case accumulation. The flag is a low-cost first move that creates the data needed to support the formal change, and the prototype demonstrates that the diagnostic-flag form is implementable rather than abstract.

E. Recommender Accountability for Synthetic Virality

The recommendation is that the systemic risk assessments Very Large Online Platforms produce under DSA Article 34 should explicitly include a synthetic-content amplification metric: the rate at which the platform's recommender system foregrounds AI-generated or AI-significantly-altered content relative to organic content, broken down by topic category and risk tier. The metric should be auditable by the European Board for Digital Services and the Commission, with comparable provisions under the UK Online Safety Act.

The basis is the "All Eyes on Rafah" virality, in which an AI-generated image accumulated approximately 45 million Instagram shares without a clear single attributable speaker (Guardian 2024); the 2024 Romanian election proceedings in which the European Commission opened formal action against TikTok over election-integrity systemic-risk failures (European Commission 2024); and the 2025 Polish complaint about TikTok-distributed AI content emphasizing "the manner in which they are distributed" (Reuters 2025b).

The primary implementation hurdle is platform opacity. The metric is meaningful only if external auditors can verify it; platforms have historically resisted feed-level data access. The recommendation should therefore couple the metric obligation with audit-access provisions analogous to those negotiated for vetted research access under DSA Article 40, so that the recommender accountability claim is verifiable and not self-reported.

VII. Limitations and Future Directions

Three limitations shape what the analysis can claim. First, news-based collection captures events that achieved media visibility and underrepresents quiet platform actions, internal enforcement, and harms in less-covered languages and regions. The 26-event dataset is a hand-coded sample, not an exhaustive catalog. Second, single-coder coding introduces interpretive variance; coding decisions are documented in the companion artifact to permit replication and disagreement, but author judgment is reflected throughout. Third, the framework-mapping exercise is interpretive: deciding whether a misfit is structural rather than patchable depends on judgments about the framework's foundational premises, judgments the present paper attempts to render transparently rather than obscure. These limitations shape the claims rather than disqualify them. The argument is that recurring, recognizable misfit patterns appear across the dataset, not that the dataset is exhaustive.

Three directions for future work follow. First, framework redesign: how should TSPA-style taxonomies absorb the actor-axis classification proposed in Recommendation D without losing operational tractability, and what governance does the proposed Hybrid-Agency category require to be more than a label? Second, longitudinal tracking: as integration deepens through 2026 and beyond, do the structural patterns identified here persist, intensify, or recede in the face of platform adaptation? Third, comparative regional analysis: how do regulatory architectures in jurisdictions less covered here, South and East Asia, Latin America, the African Union, respond to integration cases when DSA-style instruments are unavailable?

VIII. Conclusion

The argument of this paper is concrete. Trust and safety frameworks built for a world of identifiable human content creators and platform intermediaries are encountering a class of cases, platform-integrated generative AI cases above all, in which neither premise holds. A substantial proportion of dataset events still fit the existing taxonomy cleanly; the field is not in crisis. But the partial and no-fit cases cluster into recognizable structural patterns: attribution ambiguity, the collapse of the content-tool distinction, and design-layer harms with no content-level analog. Patching the taxonomy can resolve some of these. Others require the frameworks to revise the foundational premises that hold them together.

The contribution is empirical rather than programmatic. It moves the conversation from "AI creates new risks" to "here is where, why, and at what integration depth our existing classification tools become structurally inadequate," shown with cases. The five policy recommendations are responsive to those structural cases and are not intended to replace the existing regulatory architecture, which has begun moving in productive directions, but to address the gap that integration depth has opened between regulation and the institutional reality of the platforms it governs.

For the trust and safety field, the question that follows is not whether to update the TSPA taxonomy but whether the field can simultaneously hold two things: that legacy frameworks remain operationally necessary for the bulk of recognizable harms, and that integration depth has opened a category of cases for which institutional honesty about framework limits is itself part of the response. The companion artifact accompanying this paper is one attempt to make that institutional honesty inspectable, in the form of a navigable dataset and a working prototype of the diagnostic flag the paper proposes.

AI disclosure

This paper was written by the author with assistance from two AI tools used as working instruments, scoped below to keep responsibility precise.

Author work. The author defined the analytical framework (the four-axis mapping that combines TSPA Abuse Types with three non-content axes, behavioral and authenticity-based enforcement, recommendation-system intervention, and design-level safety, together with the patchable-versus-structural misfit distinction), selected the 26-event dataset from a longer candidate list of AI-related social media crises, rendered each per-event coding decision (the clean/partial/no_fit fit status across the four framework axes, the misfit_classification, and the actor_axis assignment), identified the three structural misfit patterns described in Section IV.C, conducted the engagement with counter-arguments in Section V.E, and developed the five policy recommendations in Section VI together with their implementation hurdles. The thesis, the methodology, the dataset, and the analytical judgments throughout are the author's.

Anthropic's Claude (claude-opus-4-7). The author used Claude as a working instrument for three purposes. First, literature surfacing: the author issued two structured deep-research queries (reproduced in Appendix B), one on non-content trust and safety approaches and their AI-era weakening, and one on generative AI integration into Meta, X, TikTok, and Snap. Each surfaced source was independently verified by the author against primary URLs and publication checks before inclusion; sources that could not be verified were excluded, including a reported Grok 4 launch and a Snap-specific Ofcom inquiry. Second, English-language prose support: the author developed the conceptual content, the structure, and the analytical judgments throughout, and worked with Claude on English-language rendering, sentence-level grammar and clarity improvements, prose tightening, and consistency checking across all sections. The author is a non-native English writer; AI assistance with editorial polishing was used to render the author's analytical work in clear academic English, while the substantive arguments remained the author's. Third, framework-mapping coding support: the reason-field text accompanying each event in the dataset was developed through bilingual iterative work in which the author evaluated each event's facts from Korean source materials, rendered the analytical judgment, and articulated final wording, accepting, rewriting, or rejecting candidate prose on substantive grounds.

OpenAI's ChatGPT Codex. The author used Codex to develop the companion artifact at <https://tspa-ai-mapping.vercel.app>. The author authored the structured handoff document that specified functional requirements, design constraints, and acceptance criteria; Codex assistance was scoped to site infrastructure (React components, CSS, deployment, and data-loading code). Codex did not contribute to the dataset, the framework-mapping coding, the analytical framework, the thesis, the methodology, or the policy recommendations.

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Appendix A: Event Coding Schema

Each event in the dataset is coded with the following fields:

Table 4. Coding fields used in the 26-event dataset.

Field	What it records
id, title, date, platforms, platform_family	Identification metadata for each event
ai_involvement_type	Tool / Actor / Amplifier (per Section III.B)
description	Two- to four-sentence factual summary
harm_type_primary, harm_type_secondary	TSPA harm-type assignments
actor_axis	User-Originated / User-Assisted-by-AI / Platform-Generated / Hybrid-Agency / Recommender-Amplified / Design-Layer (the parallel axis proposed in Recommendation D)
framework_mapping	Four sub-objects (tspace, behavioral, recommendation, design), each with a fit field (clean / partial / no_fit) and a reason field with a one- to two-sentence rationale
misfit_classification	clean / patchable / structural
misfit_rationale	One- to two-sentence rationale for the classification
sources	Array of citation objects (publication, title, author, date, url)

Decision rules: a fit status of *clean* requires that the framework's existing categories describe both the harm and the actor without information loss. *Partial* applies when the harm-layer fit is recognizable but the actor structure does not map cleanly, or when forced categorization loses analytically important information. *No_fit* applies when no existing category describes the event without distortion. *Patchable* misfit means the framework's foundational premises survive and a subcategory addition would resolve the gap; *structural* misfit means the foundational premises themselves are violated and revision is required.

Appendix B: Deep Research Prompts Used

The two prompts below were issued to Anthropic's Claude during literature search (per the AI disclosure section). All sources surfaced were independently verified before inclusion.

Prompt 1 (non-content T&S approaches): Surfaced foundational and practitioner literature on three non-content axes (behavioral / authenticity-based enforcement, recommendation-system intervention, design-level safety), with each source classified by its relationship to the paper's thesis (Strengthens / Tensions / Neutral) and AI-era weakening evidence requested for each axis.

Prompt 2 (platform-integrated GenAI): Surfaced documented incidents and platform governance responses for generative AI integration into Meta, X, TikTok, and Snap, with the same Strengthens / Tensions / Neutral classification and structured per-platform analysis (integration form, incidents, response, framework mapping). Both prompts can be made available to the instructor on request.

Appendix C: Companion Artifact

The companion artifact at <https://tspa-ai-mapping.vercel.app> is an interactive site that operationalizes two parts of this paper. **Section 1 (Mapping Visualizer)** provides a filterable list of all 26 dataset events. Selecting an event opens a detail panel showing the event description, source citations, the four-axis framework mapping with fit status and reason for each axis, and the misfit classification. Filters apply to year, platform family, AI involvement type, and misfit classification, with cross-filter option counts that update dynamically. **Section 2 (Hybrid-Agency Diagnostic Flag)** is a working prototype of Recommendation D. The user selects an event from the dataset or describes a new one, applies dual coding (TSPA harm-axis plus the proposed actor-axis), and receives a comparison output showing what the dual coding makes visible that single-axis coding does not.